PROJECT PROGRESS REPORT

PROJECT PROGRESS REPORT
(Reporting No. 10 / 10)
Reporting Period: From 17/08/2026 To24/09/2026

1. Student Information

Student Name	PATEL MAN VISHNUBHAI.
Enrollment Number	25IC03CA053
Program/Branch	MCA (CA)/ ICA
Semester	3
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2. Project Details

Type	Research Project
Department/Domain	MCA CA (DS/ML) /AI &ML
Mentor Name	Mr. Srikant Singh
Start Date	08-06-2026
End Date	03-10-2026

3. Project Title

- OpenCV - Based System for Crack Identification and Width Estimation

4. Progress During the Reporting Period

A. Previous Reporting Review

Planned Task	Status	Remarks
Complete Main Project Documentation	In Progress	Preparing complete project documentation, including methodology, system design, and results.
Research Paper Writing	In Progress	Working on the research paper based on project implementation and findings.
Review Research Papers and References	Completed	Reviewed relevant research papers and organized references.
Resolve Minor Project Bugs	In Progress	Identifying and fixing minor bugs to improve system stability.
Final System Testing	Completed	Tested all major modules and verified the complete workflow.
Finalize Screenshots and Result Analysis	In Progress	Organizing project screenshots and analysis results for documentation.
Final Presentation Preparation	Pending	Preparing the final presentation and project demonstration.

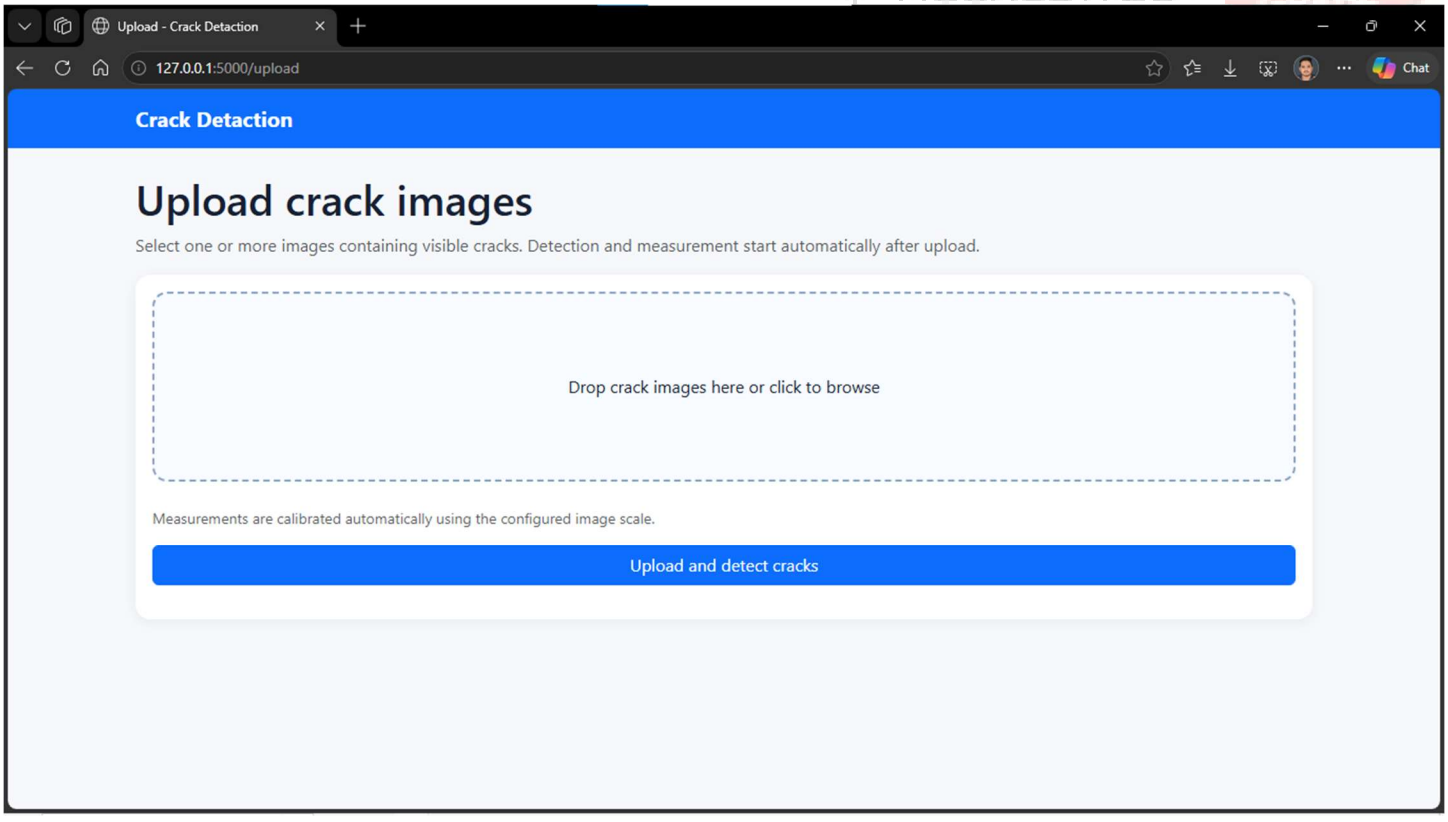
B. Work Completed During This Reporting Period

- Reviewed the complete project workflow and verified all major modules.
- Organized project results, screenshots, and generated output files.
- Reviewed research papers and organized relevant references for the research paper.
- Started preparing the main project documentation, including methodology, system design, and results.
- Identified and worked on resolving minor project bugs.
- Performed additional testing to check system stability and measurement consistency.
- Analysed crack detection, width estimation, and risk classification results.
- Continued preparing the research paper based on the project implementation and findings.

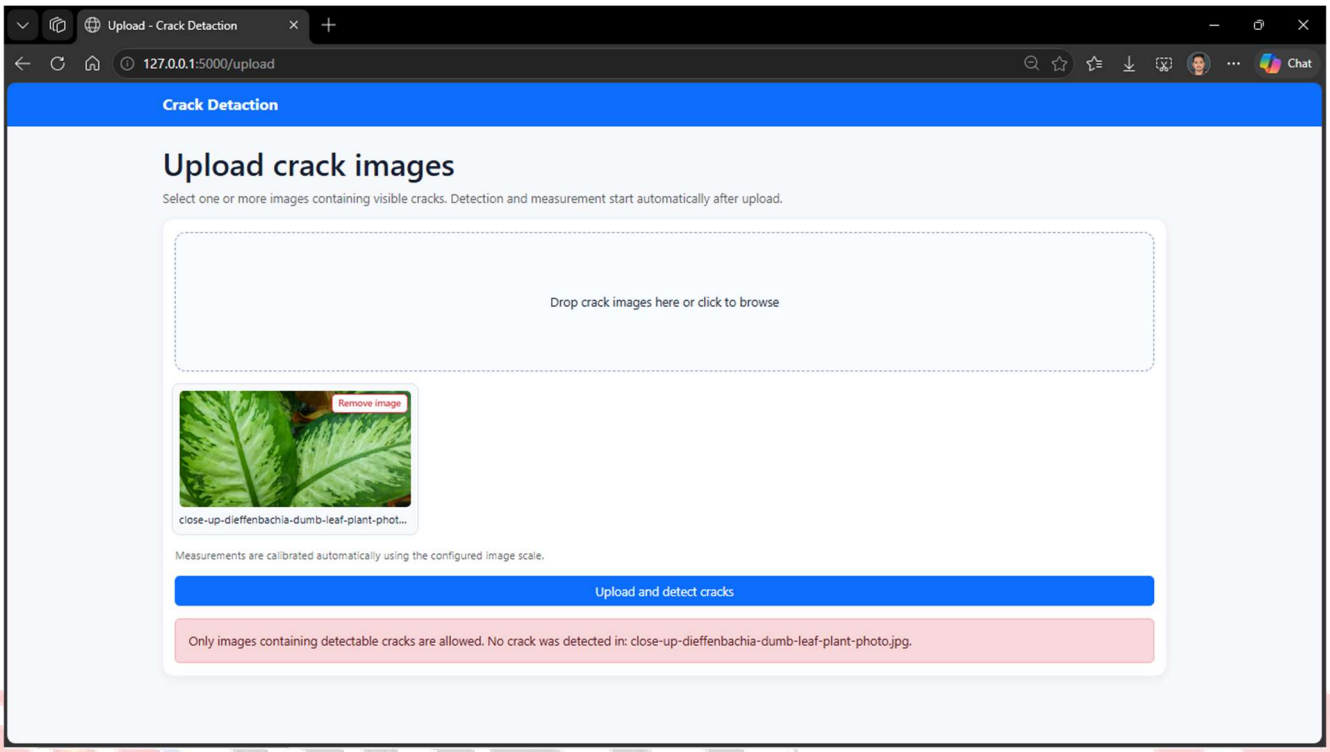
C. Evidence / Deliverables Produced

Sr. No.	Deliverable / Evidence	Description / Link / Attachment
1	Completed Project Source Code	Final Python code with all integrated crack detection, measurement, and analysis modules.
2	Screenshots and Model Output	Crack detection results, width measurements, risk classification, and annotated images.
3	CSV and ZIP Reports	Generated CSV analysis reports and ZIP packages containing project outputs.
4	Final Testing Results	Testing and validation evidence for all completed project modules.
5	Project Documentation	Documentation covering project methodology, system design, implementation, and results.
6	Research Paper Review	Reviewed research papers, literature findings, and organized references.
7	Research Paper Draft	Draft containing the research objectives, methodology, implementation, and project findings.
8	Final Presentation Materials	Organized project screenshots, results, and materials for the final presentation and demonstration.
9	Project Repository	GitHub: https://github.com/ManPatel448/OpenCV-Based-System-for-Crack-Identification-and-Width-Estimation.git

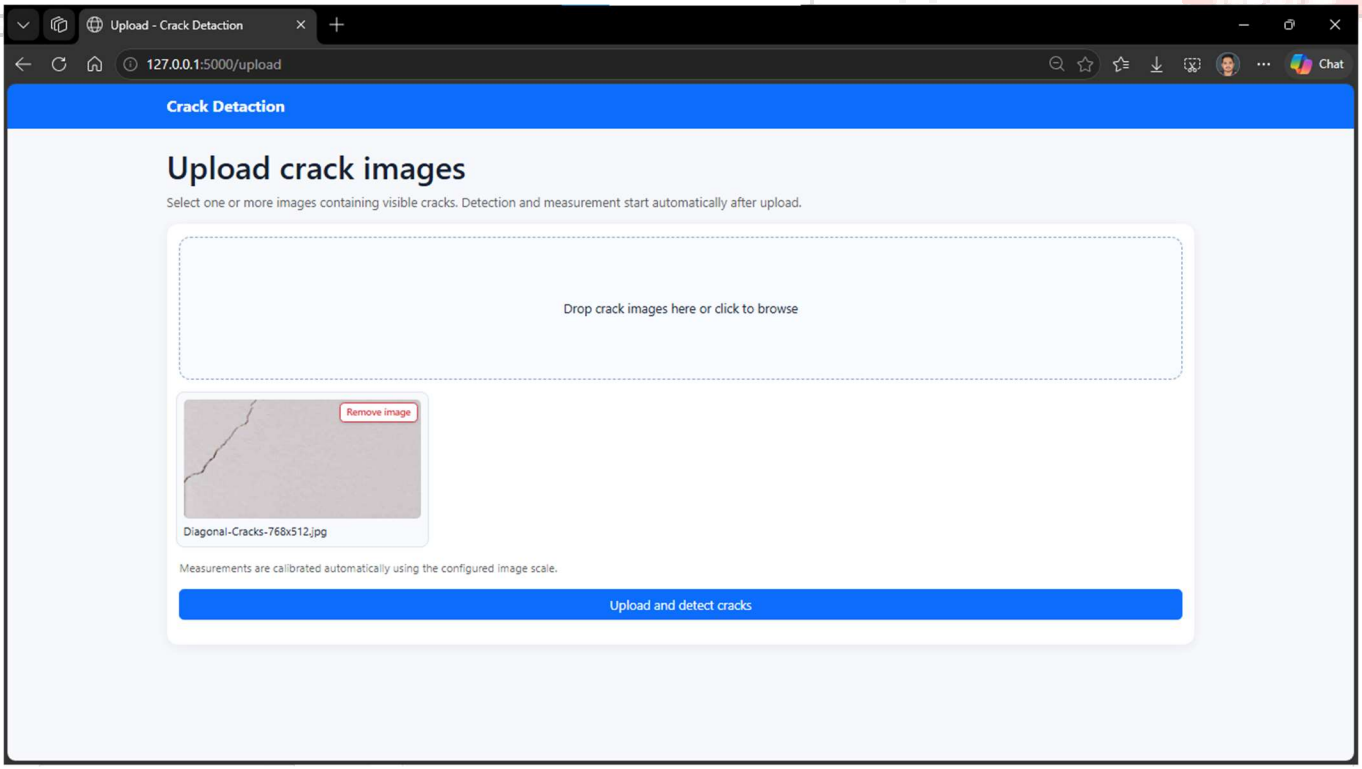
Image Upload:



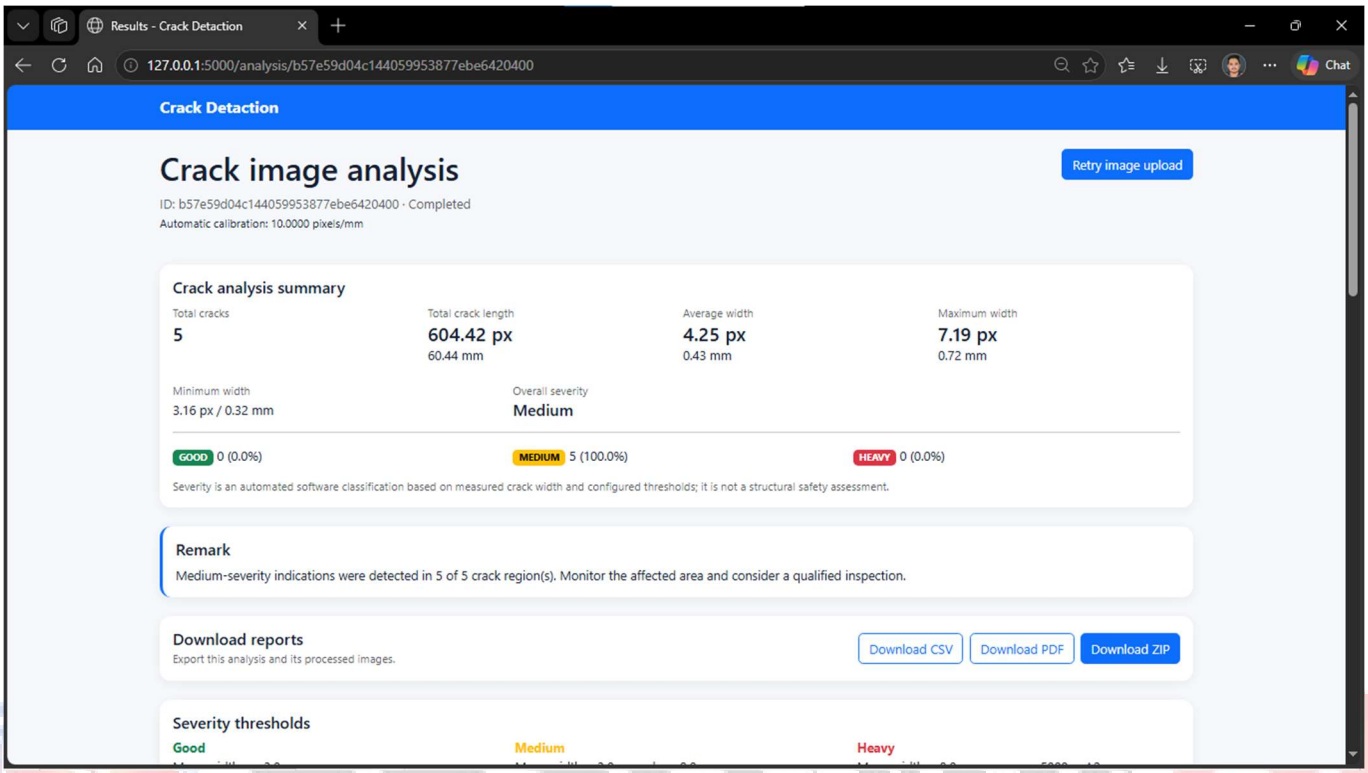
Proper Validation Only Crack Image to use:

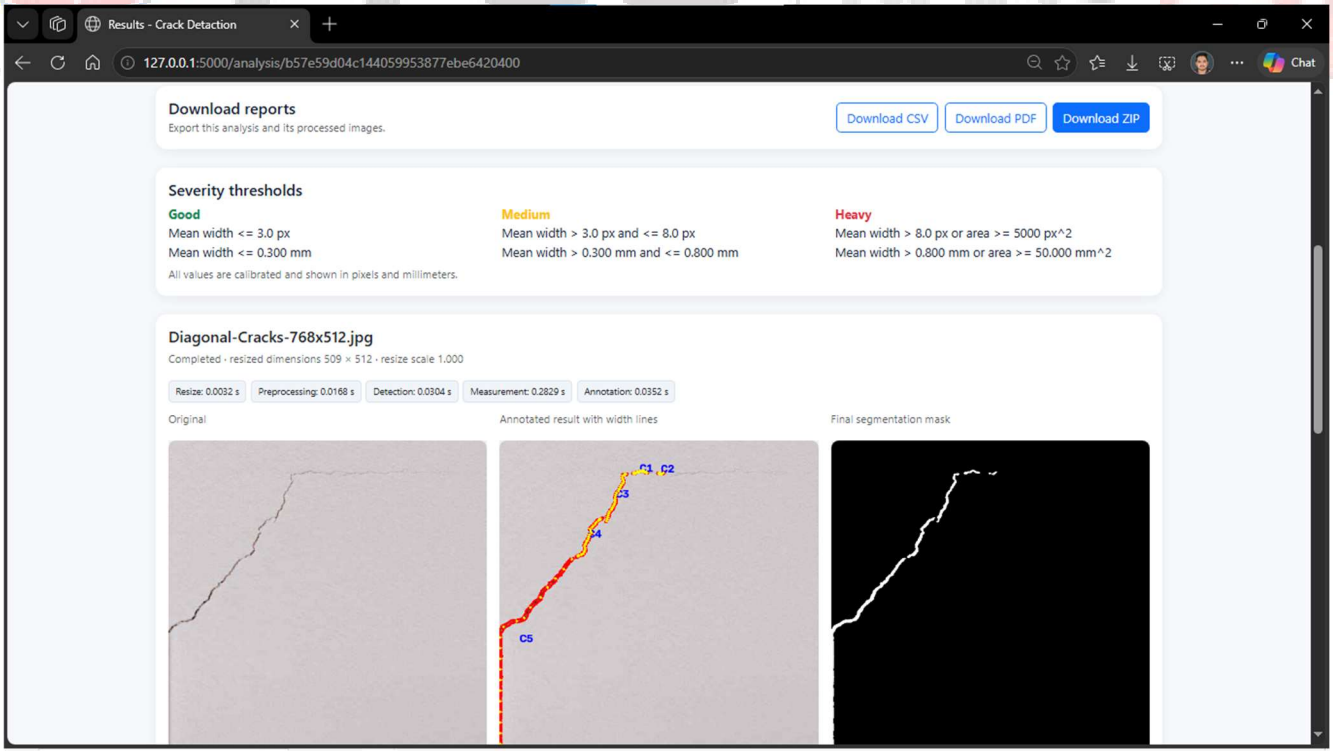


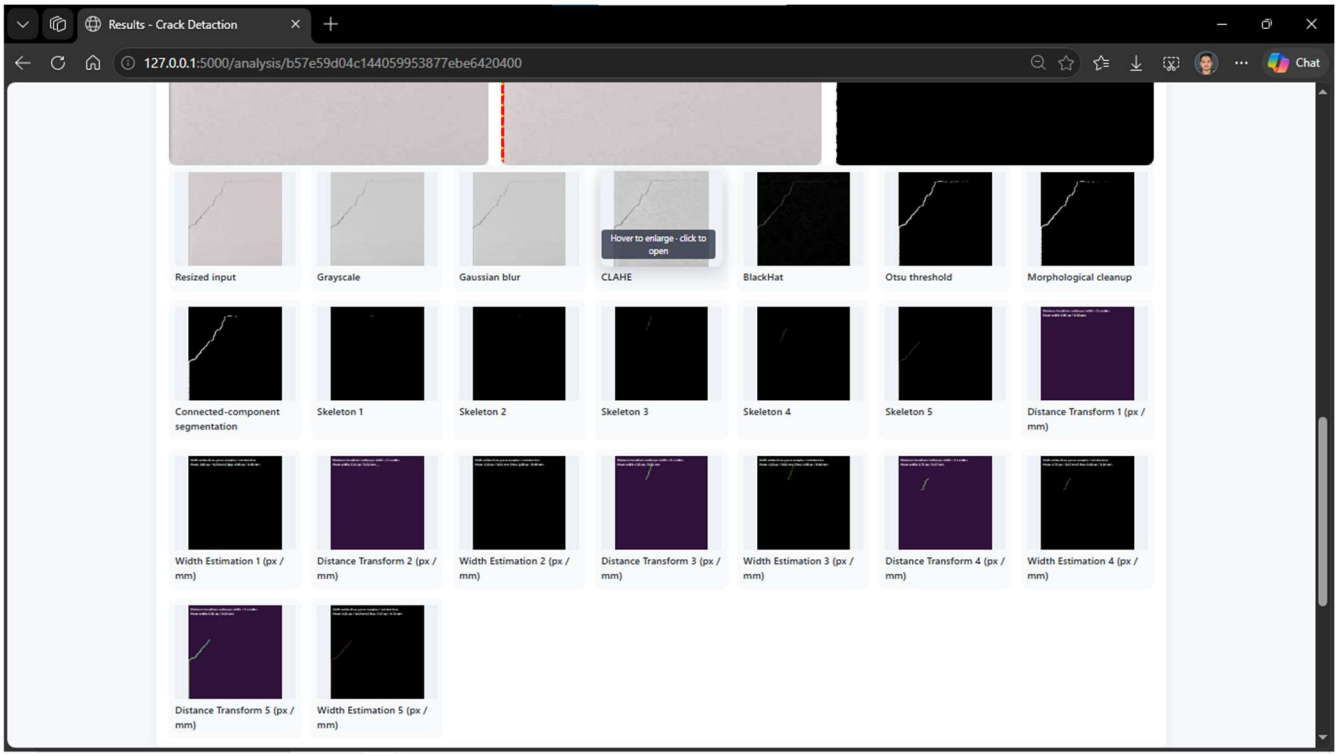
Crack Image:

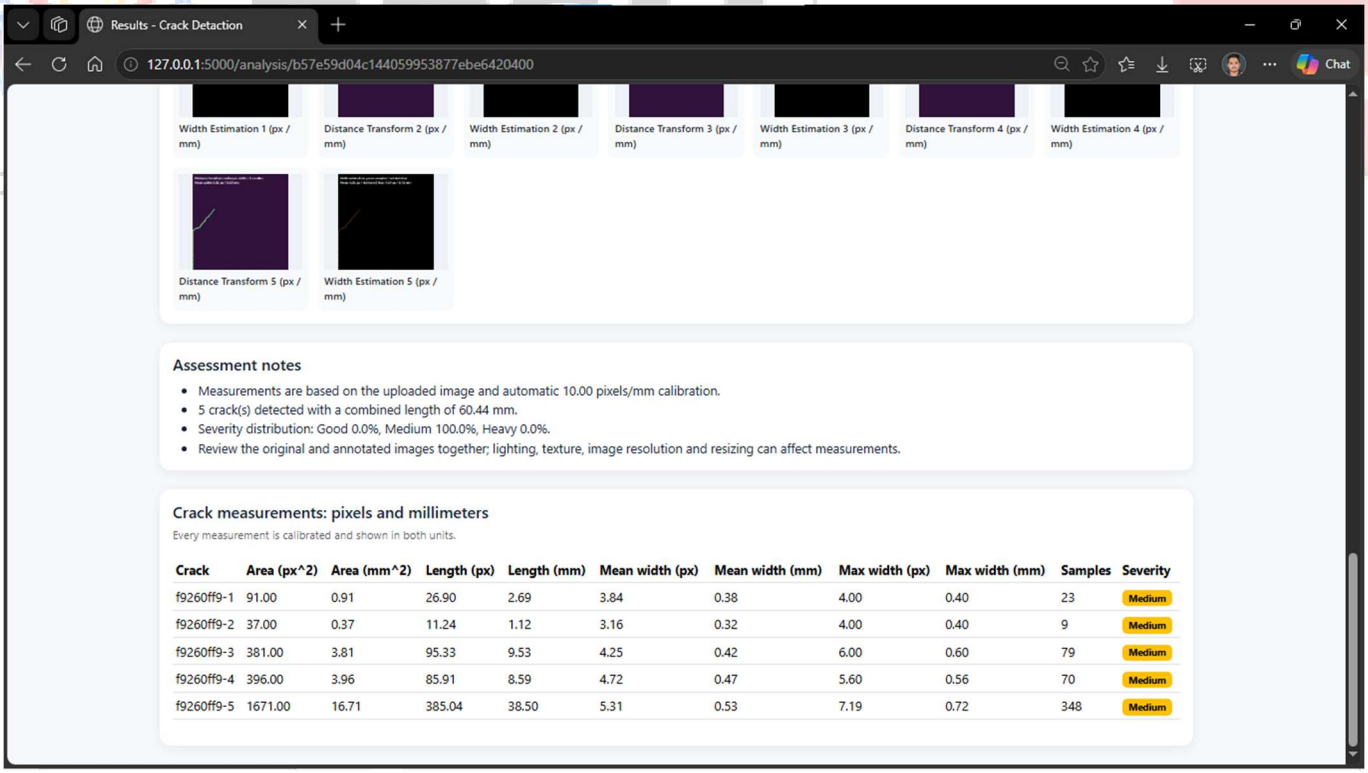


Output:


Results - Crack Detection
127.0.0.1:5000/analysis/b57e59d04c144059953877ebe6420400
Chat
Crack Detection
Crack image analysis
Retry image upload
ID: b57e59d04c144059953877ebe6420400 · Completed
Automatic calibration: 10.0000 pixels/mm
Crack analysis summary
Total cracks
5
Total crack length
604.42 px
60.44 mm
Average width
4.25 px
0.43 mm
Maximum width
7.19 px
0.72 mm
Minimum width
3.16 px / 0.32 mm
Overall severity
Medium
GOOD 0 (0.0%)
MEDIUM 5 (100.0%)
HEAVY 0 (0.0%)
Severity is an automated software classification based on measured crack width and configured thresholds; it is not a structural safety assessment.
Remark
Medium-severity indications were detected in 5 of 5 crack region(s). Monitor the affected area and consider a qualified inspection.
Download reports
Export this analysis and its processed images.
Download CSV
Download PDF
Download ZIP
Severity thresholds
Good
Medium
Heavy


Results - Crack Detection
127.0.0.1:5000/analysis/b57e59d04c144059953877ebe6420400
Chat
Download reports
Export this analysis and its processed images.
Download CSV
Download PDF
Download ZIP
Severity thresholds
Good
Mean width <= 3.0 px
Mean width <= 0.300 mm
Medium
Mean width > 3.0 px and <= 8.0 px
Mean width > 0.300 mm and <= 0.800 mm
Heavy
Mean width > 8.0 px or area >= 5000 px^2
Mean width > 0.800 mm or area >= 50.000 mm^2
All values are calibrated and shown in pixels and millimeters.
Diagonal-Cracks-768x512.jpg
Completed · resized dimensions 509 × 512 · resize scale 1.000
Resize: 0.0032 s
Preprocessing: 0.0168 s
Detection: 0.0304 s
Measurement: 0.2829 s
Annotation: 0.0352 s
Original
Annotated result with width lines
Final segmentation mask
C1 , C2
C3
C4
C5


Results - Crack Detection
127.0.0.1:5000/analysis/b57e59d04c144059953877ebe6420400
Chat
Hover to enlarge - click to open
Resized input
Grayscale
Gaussian blur
CLAHE
BlackHat
Otsu threshold
Morphological cleanup
Connected-component segmentation
Skeleton 1
Skeleton 2
Skeleton 3
Skeleton 4
Skeleton 5
Distance Transform 1 (px / mm)
Width Estimation 1 (px / mm)
Distance Transform 2 (px / mm)
Width Estimation 2 (px / mm)
Distance Transform 3 (px / mm)
Width Estimation 3 (px / mm)
Distance Transform 4 (px / mm)
Width Estimation 4 (px / mm)
Distance Transform 5 (px / mm)
Width Estimation 5 (px / mm)


Results - Crack Detection
127.0.0.1:5000/analysis/b57e59d04c144059953877ebe6420400
Chat
Width Estimation 1 (px / mm)
Distance Transform 2 (px / mm)
Width Estimation 2 (px / mm)
Distance Transform 3 (px / mm)
Width Estimation 3 (px / mm)
Distance Transform 4 (px / mm)
Width Estimation 4 (px / mm)
Distance Transform 5 (px / mm)
Width Estimation 5 (px / mm)
Assessment notes
Measurements are based on the uploaded image and automatic 10.00 pixels/mm calibration.
5 crack(s) detected with a combined length of 60.44 mm.
Severity distribution: Good 0.0%, Medium 100.0%, Heavy 0.0%.
Review the original and annotated images together; lighting, texture, image resolution and resizing can affect measurements.
Crack measurements: pixels and millimeters
Every measurement is calibrated and shown in both units.
Crack | Area (px^2) | Area (mm^2) | Length (px) | Length (mm) | Mean width (px) | Mean width (mm) | Max width (px) | Max width (mm) | Samples | Severity
f9260ff9-1 | 91.00 | 0.91 | 26.90 | 2.69 | 3.84 | 0.38 | 4.00 | 0.40 | 23 | Medium
f9260ff9-2 | 37.00 | 0.37 | 11.24 | 1.12 | 3.16 | 0.32 | 4.00 | 0.40 | 9 | Medium
f9260ff9-3 | 381.00 | 3.81 | 95.33 | 9.53 | 4.25 | 0.42 | 6.00 | 0.60 | 79 | Medium
f9260ff9-4 | 396.00 | 3.96 | 85.91 | 8.59 | 4.72 | 0.47 | 5.60 | 0.56 | 70 | Medium
f9260ff9-5 | 1671.00 | 16.71 | 385.04 | 38.50 | 5.31 | 0.53 | 7.19 | 0.72 | 348 | Medium

D. Current Project Progress

Overall Project Completion: 100%

Sr. No.	Project Component / Module	Status	Completion (%)
1	Library Installation	Completed	100%
2	Import Libraries	Completed	100%
3	Project Configuration	Completed	100%
4	Image Upload	Completed	100%
5	Image Preprocessing	Completed	100%
6	Crack Segmentation	Completed	100%
7	Crack Detection	Completed	100%
8	Crack Width Estimation	Completed	100%
9	Risk Classification	Completed	100%
10	Result Visualization	Completed	100%
11	CSV Report	Completed	100%
12	ZIP Download	Completed	100%
13	Main Project Documentation	In Progress	70%
14	Research Paper Writing	In Progress	50%
15	Minor Bug Fixes and Final Review	In Progress	80%

E. Challenges / Issues Faced

- Faced minor bugs during final project testing and system maintenance.
- Encountered challenges in organizing project results and screenshots for documentation.
- Faced difficulties in structuring the research paper and presenting technical findings clearly.
- Required additional verification of crack-width measurements and risk classification results.
- Needed to organize research references and maintain proper citation formats.
- Faced challenges in ensuring consistency between project outputs, analysis, and documentation.

F. Actions Taken / Solution Implemented

- Identified and resolved minor bugs to improve system stability.
- Organized project screenshots, output files, and analysis results for documentation.
- Reviewed crack detection and width estimation results for accuracy and consistency.
- Structured the main project documentation with methodology, system design, and results.
- Organized research papers, references, and citations for the research paper.
- Improved the presentation of technical findings and project outputs.
- Performed additional testing and verification of the completed system modules.

G. Plan for Next Reporting

- Complete the main project documentation with final results and analysis.
- Finalize the research paper writing and review its content.
- Resolve remaining minor bugs and perform final system verification.
- Organize final project screenshots, diagrams, and output evidence.
- Verify all references, citations, and research findings.
- Prepare the final project presentation and demonstration.
- Complete the final review and prepare all project files for submission.

H. Learning / Skills Acquired

- Improved Python and OpenCV skills through final testing and project maintenance.
- Gained practical experience in crack detection, width estimation, and result analysis.
- Enhanced debugging, troubleshooting, and problem-solving skills.

- Improved technical documentation and research paper writing skills.
- Learned research analysis, reference management, and citation formatting.
- Strengthened testing, validation, and quality assurance skills.
- Improved technical communication and project presentation skills.

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5. Tasks Accomplished

- Completed final testing and verification of all major project modules.
- Organized project screenshots, output files, and crack analysis results.
- Reviewed research papers and finalized relevant references.
- Prepared the main project documentation structure and technical content.
- Analyzed crack detection, width estimation, and risk classification results.
- Identified and resolved minor project bugs.
- Improved system stability and verified measurement consistency.
- Continued research paper writing based on project implementation and findings.

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6. Tasks in Progress / Next Plan

- Complete the main project documentation with final results and analysis.
- Finalize the research paper with proper references and citations.
- Resolve any remaining minor project bugs and perform final verification.
- Organize final screenshots, diagrams, and project output evidence.
- Review and refine the research findings and technical content.
- Prepare the final project presentation and demonstration.
- Complete the final review and prepare all project files for submission.

7. Learnings / Skills Acquired

- Improved Python programming and OpenCV skills through project development and testing.
- Gained practical knowledge of crack detection, segmentation, and width measurement.
- Enhanced debugging, troubleshooting, and system maintenance skills.
- Developed technical documentation and research paper writing skills.
- Improved research analysis, reference management, and citation formatting.
- Strengthened testing, validation, and result analysis skills.
- Improved technical communication, presentation, and project management skills.

References:

- Li, Y. (2026). Crack Detection and Semantic Segmentation in Concrete Structures Using an Improved Deep Learning-Based Framework. Journal of Engineering and Applied Science, 73, 107.
- Grzesiak, S., Nogatz, T., de Sousa, C., Kronenberger, M., Redenbach, C., & Pahn, M. (2026). In Situ 4D Image Analysis of Crack Width and Strain Changes in GFRP-Reinforced Concrete. Materials and Structures, 59, 197.
- Hassouna, M., Marzouk, M., & Fathalla, E. (2026). Automated Low-Cost Framework for Crack Measurements in RC Structures Using Deep Learning Approach. Scientific Reports, 16, 14678.
- Peng, S., Zhao, B., Cao, J., et al. (2026). A Review of Machine Learning-Based Methods for Concrete Crack Detection. Journal of Intelligent Construction, 4(2), 9180121.

Evaluation (To be filled by Mentor)

Criteria	Bloom's Level	Excellent (20)	Good (15)	Average (10)	Poor (5)	Marks
Progress as per Previous Plan	L3– Apply	All planned tasks completed	Most completed	Partial completion	Very little progress	
Technical / Research Implementation	L4– Analyze	Excellent implementation with clear evidence	Good implementation	Basic implementation	Insufficient evidence	
Quality of Deliverables	L5– Evaluate	High-quality, validated deliverables	Good quality	Acceptable	Poor / incomplete	
Problem Solving & Challenges	L4– Analyze	Challenges clearly identified and effectively solved	Most issues addressed	Limited solutions	No clear effort	
Planning for Next Phase	L6– Create	Clear, realistic and measurable plan	Mostly clear plan	Basic plan	No proper plan	
Documentation & Presentation	L3– Apply	Well-structured and professional	Minor issues	Average clarity	Poor documentation	
Total marks out of 100						

Evaluation

Student Signature	Mentor (Institute) Signature

Mentor (Institute) Remarks

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